

Here are **scenario-based constructor chaining tasks** that require students to think about the design instead of simply following constructor signatures.

Task 1: Hospital Patient Registration System

Scenario

A hospital is developing a patient registration system.

When a patient first visits the hospital, only a **Patient ID** is generated.

Later, the receptionist adds the **Patient Name**.

After the doctor examines the patient, the **Disease** is updated.

Finally, the billing department records the **Bill Amount**.

As a Java developer, design the **Patient** class so that every stage of registration reuses the previous stage's initialization instead of writing the same code repeatedly.

Requirements

- Create a **Patient** class.
 - Use **constructor chaining** (**this()**).
 - Do not duplicate initialization code.
 - Create an object with all patient details.
 - Display all patient information.
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Task 2: Online Food Delivery Application

Scenario

An online food delivery application allows customers to place an order in multiple steps.

Initially, the system creates an order with only the **Order ID**.

After the customer selects a restaurant, the **Restaurant Name** is added.

Next, the customer chooses the **Food Item**.

Finally, the customer enters the **Delivery Address** before placing the order.

As a Java developer, implement the `FoodOrder` class so that each stage of order creation reuses the previous constructor instead of repeating the same initialization code.

Requirements

- Create a `FoodOrder` class.
- Use **constructor chaining** (`this()`).
- Avoid duplicate code.
- Create an object with complete order details.
- Display all order information.

Since the tasks don't provide actual code, there is no fixed output. Assuming the constructors are chained correctly and an object is created with all the details, the expected output would be similar to the following.

Task 1: Hospital Patient Registration System

If the object is created as:

Java

```
Patient p = new Patient(101, "Rahul", "Fever", 2500);
```

Expected Output

None

```
Patient Registration Started  
Patient ID : 101  
Patient Name : Rahul  
Disease : Fever  
Bill Amount : 2500.0
```

Task 2: Online Food Delivery Application

If the object is created as:

Java

```
FoodOrder f = new FoodOrder(1001, "Paradise", "Chicken Biryani",  
"Hyderabad");
```

Expected Output

None

```
Food Order Started  
Order ID : 1001  
Restaurant : Paradise  
Food Item : Chicken Biryani  
Delivery Address : Hyderabad
```